

Yi Xia (Sean)

Applied AI Engineer • Computer Vision • End-to-end ML Deployment

Tāmaki Makaurau Auckland • Aotearoa
New Zealand

+64 27 368 7961

xiayi.shawn.001@gmail.com

linkedin.com/in/yi-xia • Google Scholar

Open Work Rights • Post-Study Work Visa

Applied AI engineer who ships **computer-vision systems to production**. Four years at **TOHU MEDIA**, a Māori-owned media-production company, designing and deploying object-detection and OCR pipelines end-to-end — problem framing, dataset spec, PyTorch training, **Docker-packaged inference on cloud GPU servers**, and iteration with end users on what "useful output" looks like inside a real workflow. Concurrently completing a PhD at **Auckland University of Technology** on deep learning for video understanding; first-author paper accepted at **Pattern Recognition** (Q1, 2026), six peer-reviewed papers, 70+ citations. Looking for a senior **Applied AI / Computer Vision / Machine Learning** role with a New Zealand organisation that needs CV translated from research into reliable production systems.

TECHNICAL SKILLS

Production & Deployment

Docker • containerised model inference on cloud GPU servers • full ML lifecycle ownership • latency ↔ accuracy trade-offs

Computer Vision

Object detection (incl. YOLO family) • object tracking • OCR • action recognition • semantic / instance segmentation • classification • video analytics

Deep Learning

PyTorch (primary) • TensorFlow / Keras • OpenCV • scikit-learn • NumPy • pandas • Hugging Face

Programming

Python (expert) • Java

Generative AI & LLM

LLM integration • agentic workflows • agent skills & tool use • Claude / GPT APIs • prompt engineering

Research methods

Graph convolutional networks • attention & Transformers • skeleton-based activity recognition • frequency-domain learning

Tooling

Git / GitHub • Linux • CUDA • Jupyter

DOMAIN EXPOSURE

- **Media / film production** — current TOHU MEDIA work: object detection & OCR on long-form video
- **Agriculture / horticulture** — published research on kiwifruit detection & tracking from orchard video (50+ citations)
- **Activity recognition** — PhD line on skeleton-based human action understanding; applications across healthcare, sports, safety and elder-care monitoring

EDUCATION

PhD, Computer & Information Sciences

Auckland University of Technology

Oct 2023 – Oct 2026 (expected)

Deep learning for skeleton-based human activity recognition.

AUT PhD Full Scholarship.

MSc, Computer & Information Sciences

Auckland University of Technology

2021 – 2023 • Specialisation: Computer Vision

BEng, Logistics Engineering (Software)

Dalian Neusoft University of Information, China

2011 – 2015

SELECTED EXPERIENCE

Artificial Intelligence Developer at TOHU MEDIA

Auckland • Feb 2022 – present

Full-time since June 2025; previously part-time alongside MSc and PhD

- **Own the company's AI capability end-to-end** at a Māori-owned media-production company — design and ship object-detection and OCR systems that operate on client video footage inside internal production pipelines (specific deliverables under NDA).
- Built CV systems that **work on real production footage**, not just clean benchmark data — handled variable shot conditions, dense overlay text in non-standard layouts, archival material with poor contrast, and long-form takes that exceed single-workstation GPU memory. Selected model architecture, training regime and post-processing accordingly.
- Took every system through the **full ML lifecycle**: dataset scope and labelling spec, PyTorch training and evaluation, **Docker-packaged inference deployed to the company's cloud GPU servers**, and iteration with editors and the creative director on what "useful output" means inside an actual editing workflow.
- Made independent calls on architecture, **latency / accuracy trade-offs** and scope reduction under deadline — four years part-time alongside the MSc and PhD, now full-time ownership of the AI work.

PhD Researcher at Institute of Robotics & Vision, AUT

Oct 2023 – Oct 2026

Supervisors: Dr Raymond Lutui (primary) · Assoc. Prof. Sira Yongchareon

- Designed a **frequency-aware spatio-temporal graph network** for skeleton-based human activity recognition; outperforms prior state-of-the-art on standard benchmarks — accepted at *Pattern Recognition* (Q1 journal, IF = 7.6) in 2026.
- Authored a peer-reviewed paper at **ICONIP 2024** on dynamic self-attention gating between spatial and temporal streams in video understanding.
- **Full ownership of first-author research** — idea, model design, GPU training, ablations, paper writing and reviewer rebuttal.
- Mentor master's students on PyTorch, experimental hygiene and academic writing.

Research Assistant at School of Engineering, Computer & Mathematical Sciences, AUT

Nov 2022 – Jul 2023

- Funded research assistant on the **kiwifruit detection & counting** programme — co-built KiwiDetector (improved YOLOv7) and KiwiTracker for real-time orchard video. The lead paper now has **50+ citations**.
- Co-authored three peer-reviewed papers (IVCNZ 2022, SAI 2023, PSIVT 2023) during the assistantship.

Teaching Assistant at AUT

2022 – present

- TA on multiple undergraduate computer-science courses (programming, computer vision, AI). Run labs, mark assignments, hold consultation hours and mentor on final-year projects.

Tech Service Specialist at PB Tech

Auckland • Apr – Dec 2023

- Diagnosed hardware and software issues for walk-in customers in a busy retail tech-service environment — sharpened plain-English technical communication with non-specialist users.

Co-Founder • Junior Lecturer at Baiyin Pinuo Training School · Baiyin Vocational & Technology College

China • 2015 – 2021

- **Junior Lecturer (2015 – 2019)**: taught Computer Fundamentals, Java, IoT applications and RFID technologies to ~120 students per year.

- **Co-Founder (2019 – 2021):** co-founded a private training school; built curriculum, ran operations and finance; divested in 2021 to relocate to New Zealand for postgraduate study — the school remains in operation under my co-founder.

SELECTED PUBLICATIONS

6 papers · 70+ citations · h-index 3

Y. Xia, S. Yongchareon, R. Lutui, Q. Z. Sheng (2026). “Frequency-Aware Spatio-Temporal Topology Learning for Skeleton-Based Human Activity Recognition.” *Pattern Recognition*. **Q1 journal**
doi.org/10.1016/j.patcog.2026.113146

Y. Xia, M. Nguyen, W. Q. Yan (2022). “A Real-Time Kiwifruit Detection Based on Improved YOLOv7.” *IVCNZ 2022*, Springer LNCS. **50 citations**
doi.org/10.1007/978-3-031-25825-1_4

Y. Xia, S. Yongchareon, R. Lutui, Q. Z. Sheng (2025). “Dynamic Self-Attention Gated Spatial-Temporal GCN for Skeleton-Based HAR.” *ICONIP 2024*, Springer LNCS.
doi.org/10.1007/978-981-96-6588-4_14

+ three further peer-reviewed papers on kiwifruit detection & tracking — full list on [Google Scholar](#).

REFERENCES

Available on request. Doctoral supervisors (Dr Raymond Lutui, AUT; Assoc. Prof. Sira Yongchareon, AUT) and an industry referee from TOHU MEDIA can be provided.